

Sistemul numelor de domenii

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Cuprins

- Domain Name System (DNS)
 - Caracterizare
 - Organizare
 - Configurare
 - Comenzi, Primitive
 - Internationalized Domain Name (IDN)

DNS

- Adresele IP (de ex. 85.122.23.145, 2001:0db8:0001:0000:0000:0ab9:C0A8:0102) sunt dificil de memorat
- Se utilizeaza un **sistem al numelor de domenii** pentru a translata adresele IP in nume de domenii si invers
- Numele de domenii se organizeaza in ierarhii
- DNS este definit in numeroase documente RFC (e.g., 1034, 1035, 1123, 2181, 5890, 5891 etc.)
- www.statdns.com/rfc/

DNS | organizare

- Initial: **/etc/hosts** – perechi (nume, IP)
– probleme de scalabilitate
- Actual: DNS consta dintr-o schema ierarhica de nume de domenii si dintr-un sistem de baze de date distribuite ce implementeaza aceasta schema de nume

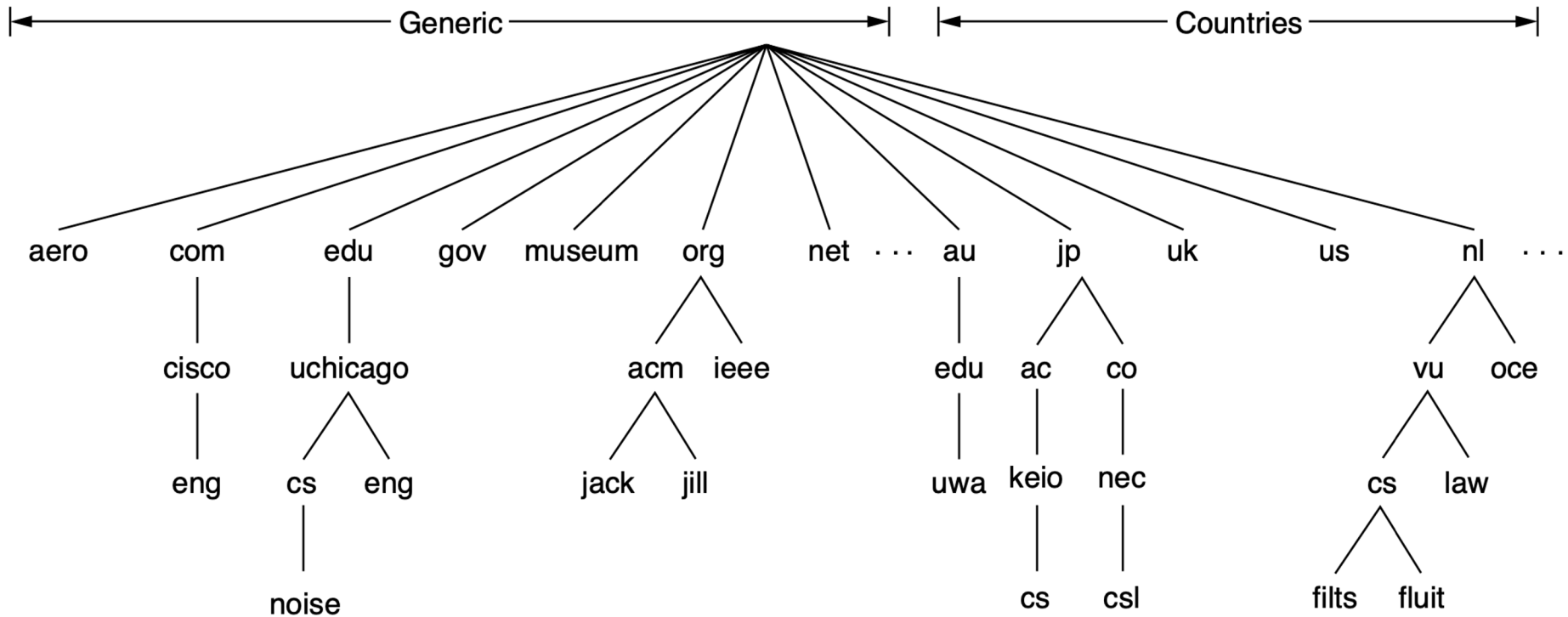


Figura. O portiune a spatiului numelor de domenii in Internet

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

DNS | Tipuri de domenii

- **Primare** (*Top Level Domains* – TLDs)
 - pentru Infrastructura Internet – un singur domeniu **.arpa** ARPA (Address and Routing Parameter Area)
 - administrat de IANA
 - a fost primul TLD
 - utilizat initial pentru tranzitia catre DNS
 - in prezent utilizat in anumite contexte particulare (de ex. *reverse DNS resolution*)
 - State (*ccTLD*) – coduri de state: .ro, .fr, .jp, ... (definite in ISO 3166)
 - Generice: .biz, .com, .info, .name, .net, .org, .pro, ...
 - in 2011 existau doar 22 gTLDs
 - din 2011, ICANN a permis inregistrarea unor gTLDs personalizate (costul initial pentru aplicare era de 200.000 USD)

DNS | Tipuri de domenii

- **Primare** (*Top Level Domains* – TLDs)
 - Sponsorizate: .aero, .edu, .gov, .int, .jobs, .mil, .tel, ...
 - Rezervate: .example, .invalid, .localhost, .test, ... (RFC 2606)
 - nu sunt expuse public, in Internet
 - utilizate pentru teste locale, in documentatii etc.
 - in iulie 2024 ICANN a aprobat un TLD nou, .internal
 - rezervat utilizarii in scop privat, in retele interne (asa cum exista anumite clase de IP-uri locale)
 - nu va fi delegat in radacina DNS
 - scopul este de a standardiza utilizarea unui TLD in retele interne, incat diferite organizatii sa nu isi mai creeze ad-hoc propriile TLDs, asa cum au procedat pana in prezent
 - <https://www.icann.org/en/board-activities-and-meetings/materials/approved-resolutions-special-meeting-of-the-icann-board-29-07-2024-en#section2.a>

DNS | Tipuri de domenii

- **Primare** (*Top Level Domains – TLDs*)
 - IDN TLDs (*Internationalized Top-Level Domains*)
 - http://example.test -> http://παράδειγμα.δοκιμή (Greek) (experimental)
 - http://example.test -> http://例え.テスト (Japanese) (experimental)
 - http://example.test -> http://उदाहरण.परीक्षा (Hindi) (experimental)
 - .中国 (ccTLD for China)
 - .닷넷 , .닷컴 (gTLDs in South Korea, meaning “dotnet” and “dotcom”)
 - Pseudo-domenii: .bitnet, .local, .root, .uucp, .onion, .crypto, .bitcoin, .eth ...
 - nu fac parte din sistemul DNS oficial

DNS | Tipuri de domenii

https://www.iana.org/domains/root/db



Domain Names

Overview

Root Zone Management

Overview

Root Database

Hint and Zone Files

Change Requests

Instructions & Guides

Root Servers

.INT Registry

.ARPA Registry

IDN Practices Repository

Root Key Signing Key (DNSSEC)

Reserved Domains

Root Zone Database

The Root Zone Database represents the delegation details of top-level domains, including gTLDs such as .com, and country-code TLDs such as .uk. As the manager of the DNS root zone, we are responsible for coordinating these delegations in accordance with our [policies and procedures](#).

Much of this data is also available via the WHOIS protocol at whois.iana.org.

DOMAIN	TYPE	TLD MANAGER
.aaa	generic	American Automobile Association, Inc.
.aarp	generic	AARP
.arth	generic	Fiat Chrysler Automobiles N.V.
.abb	generic	ABB Ltd
.abbott	generic	Abbott Laboratories, Inc.
.abbvie	generic	AbbVie Inc.
.abc	generic	Disney Enterprises, Inc.
.able	generic	Able Inc.
.abogado	generic	Minds + Machines Group Limited
.abudhabi	generic	Abu Dhabi Systems and Information Centre
.ac	country-code	Network Information Center (AC Domain Registry) c/o Cable and Wireless (Ascension Island)
.academy	generic	Binky Moon, LLC
.accenture	generic	Accenture plc

DNS | Tipuri de domenii

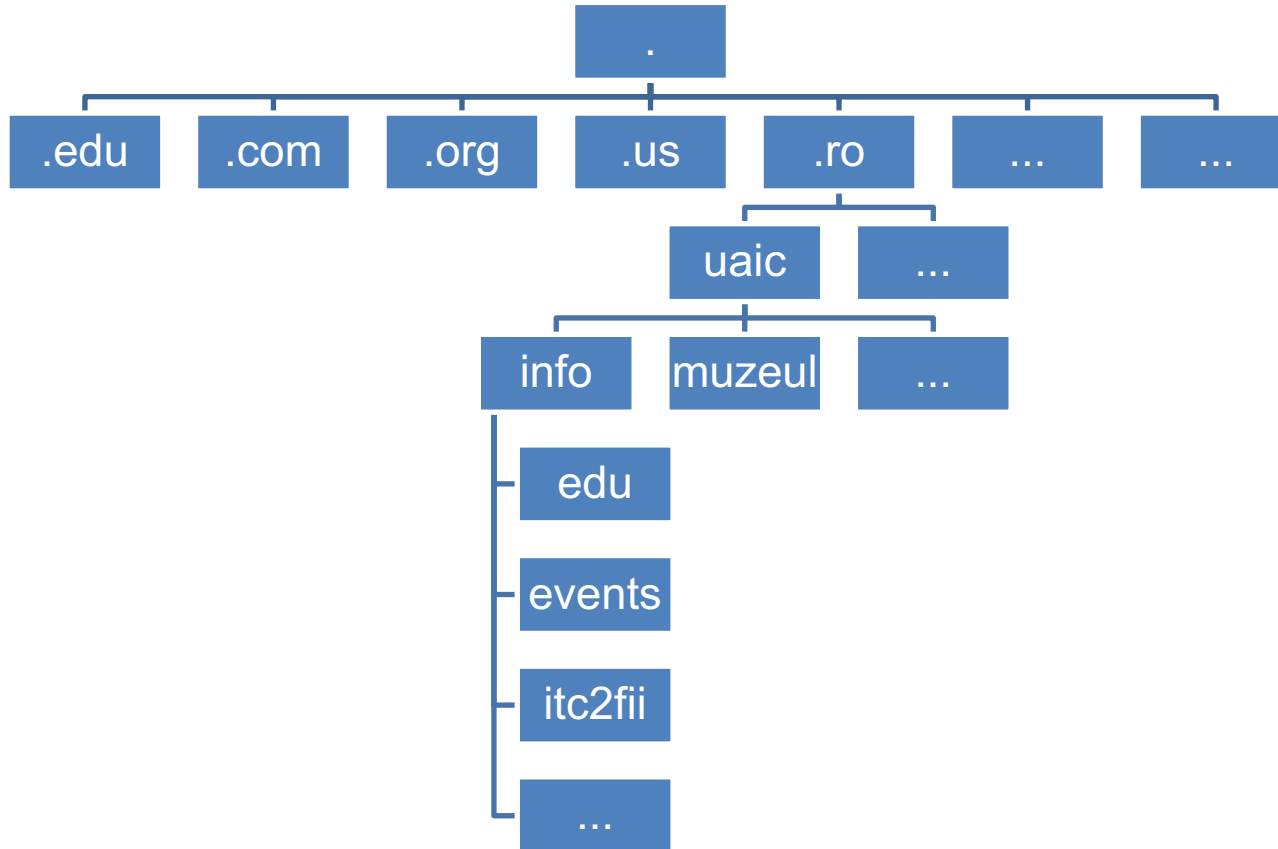
- Domeniu de nume
 - Subarbore al arborelui de domenii
 - Nu trebuie sa respecte topologia rețelei fizice
- **Sub-domenii**
- **Nume de calculatoare (gazde)**

! intreaga cale de nume nu poate depasi 255 de caractere

! numele unei componente nu poate depasi 63 de caractare

DNS

- Exemplu:



DNS | organizare

- Reguli de alocare a numelor de domenii:
 - Fiecare domeniu controleaza cum sunt alocate subdomeniile sale
 - Pentru a crea un nou subdomeniu, se cere permisiunea domeniului in care va fi inclus (un domeniu de la un anumit nivel va avea o autoritate)
 - Atribuirea de nume de domenii respecta granitele organizationale, nu pe cele ale retelelor
 - Un anumit nivel din ierarhia de niveluri poate fi controlat de mai multe servere

DNS | organizare

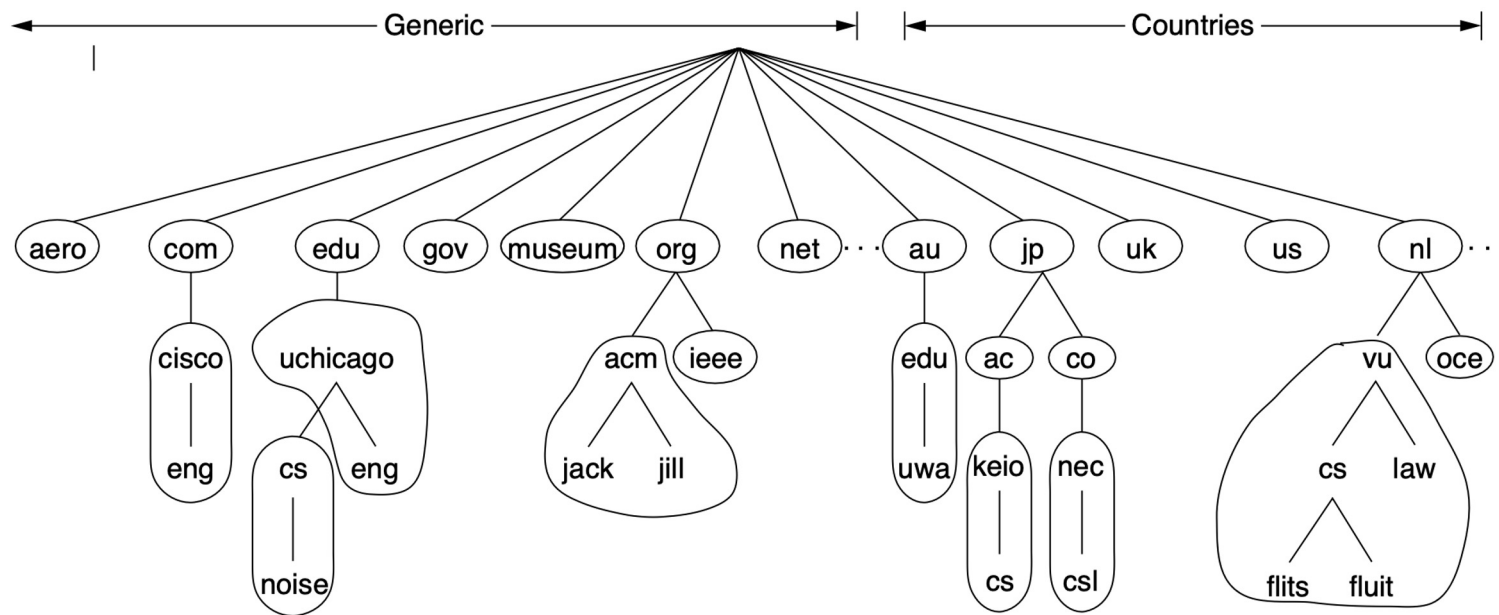
- Servere de nume (*name servers*)
 - Teoretic, un singur server de nume poate contine intreaga baza de date DNS si poate raspunde tuturor cererilor
 - Probleme: incarcarea si “*single point of failure*”
- Spatiul de nume DNS se divide in zone nesuprapuse



DNS | organizare

- Servere de nume (*name servers*)

Exemplu: O posibila impartire a spatiului de nume DNS in zone



[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

DNS | organizare

- Servere de nume (*name servers*)
 - exista un **server primar** (*primary/authoritative name server*) care deserveste un anumit domeniu si, eventual, mai multe **servere secundare** continand baze de date replicate
 - TCP se utilizeaza pentru replicarea DNS
 - uzual, UDP pentru interogari (*lookups*)
 - inclusiv in noul standard DNS-over-QUIC (DoQ) (RFC 9250)

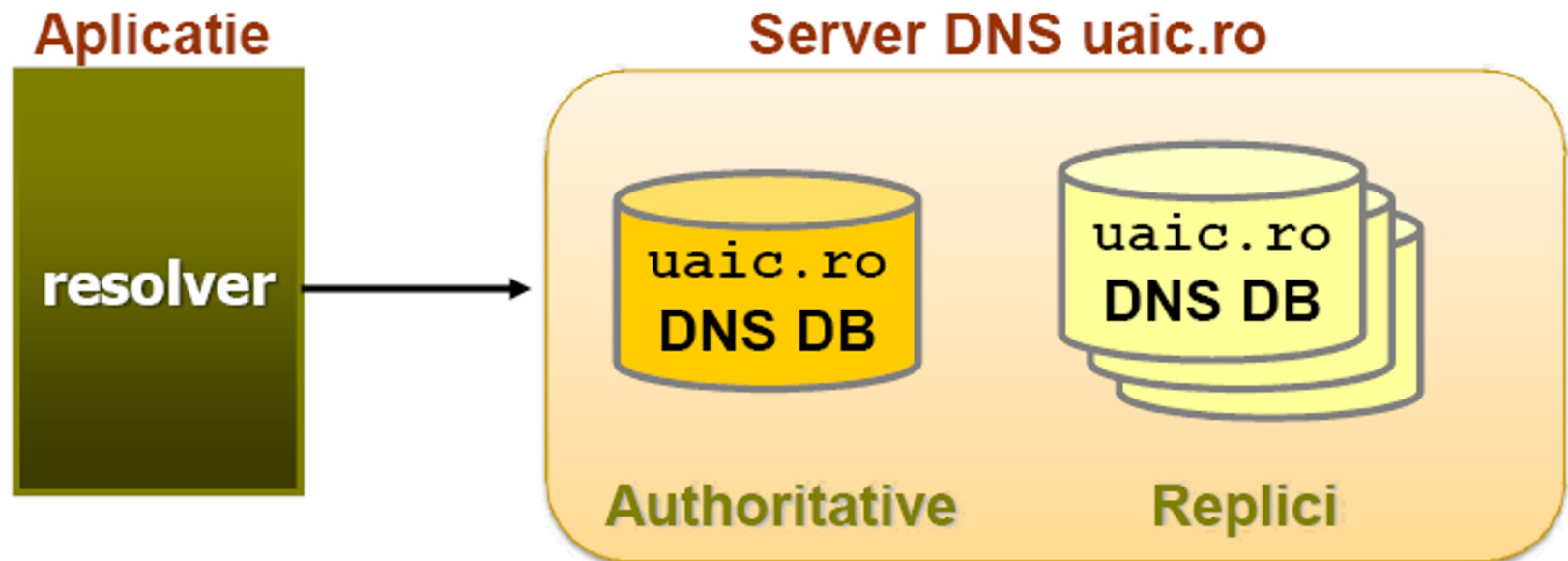
! In prezent, si TCP este utilizat pentru efectuarea interogarilor, din considerente de securitate (a se vedea DNSSEC (RFC 9364), DNS-over-TLS (DoT) (RFC 7858), DNS-over-HTTPS (DoH) (RFC 8484))

blog.apnic.net/2022/09/02/doh-dot-and-plain-old-dns/

stats.labs.apnic.net/edns

DNS | organizare

- Client DNS
 - Denumit **resolver**, trimite un pachet UDP serverului DNS, care cauta numele si returneaza adresa IP



DNS | organizare

- Exemplu de implementari ale serverului de nume: **BIND** (Berkeley Internet Name Domain), CoreDNS, Microsoft DNS Server, PowerDNS etc.
- Ca resolver (client) interactiv, se poate folosi de exemplu una dintre comenzile: **nslookup**, **host** sau **dig**.

DNS | interogari

- Interogari:
 - **Recursiva** – daca un server DNS nu cunoaste adresa pentru numele solicitat, atunci va interoga alt server DNS
 - **Incrementala** – daca serverul DNS nu stie sa raspunda, returneaza eroare si adresa altui server DNS (numit si *referral*) care ar putea cunoaste raspunsul la interogare

[learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2003/cc775637\(v=ws.10\)](https://learn.microsoft.com/en-us/previous-versions/windows/it-pro/windows-server-2003/cc775637(v=ws.10))

DNS | interogari

- Fiecarui domeniu ii este asociata o multime de inregistrari de resurse (**resource record – RR**)
- Mecanismul:
 - Cererea: *resolver*-ul trimite un nume de domeniu
 - Raspunsul: inregistrarile de resurse asociate acelui nume (stocate in bazele de date DNS)



DNS realizeaza corespondenta dintre numele de domenii si inregistrarile de resurse

DNS | interogari

- Forma generala RR este:

Nume_domeniu Timp_de_viata Clasa Tip Valoare

Nume_domeniu (*domain name*) – precizeaza domeniul caruia i se aplica aceasta inregistrare

Timp_de_viata (*time_to_live*) – da o indicatie despre cat de stabila este inregistrarea

Clasa – pentru Internet valoarea acestuia este IN

Tip – specifica tipul inregistrarii

Valoare – acest camp poate fi un numar, un nume de domeniu sau un sir ASCII; semantica depinde de tipul de inregistrare

DNS | interogari

Tabel. Tipurile principale de inregistrari DNS

Type	Meaning	Value
SOA	Start of authority	Parameters for this zone
A	IPv4 address of a host	32-Bit integer
AAAA	IPv6 address of a host	128-Bit integer
MX	Mail exchange	Priority, domain willing to accept email
NS	Name server	Name of a server for this domain
CNAME	Canonical name	Domain name
PTR	Pointer	Alias for an IP address
SPF	Sender policy framework	Text encoding of mail sending policy
SRV	Service	Host that provides it
TXT	Text	Descriptive ASCII text

[Computer Networks, 2021,
Andrew S. Tanenbaum et. al.]

DNS | interogari

Tip - precizeaza tipul inregistrarii

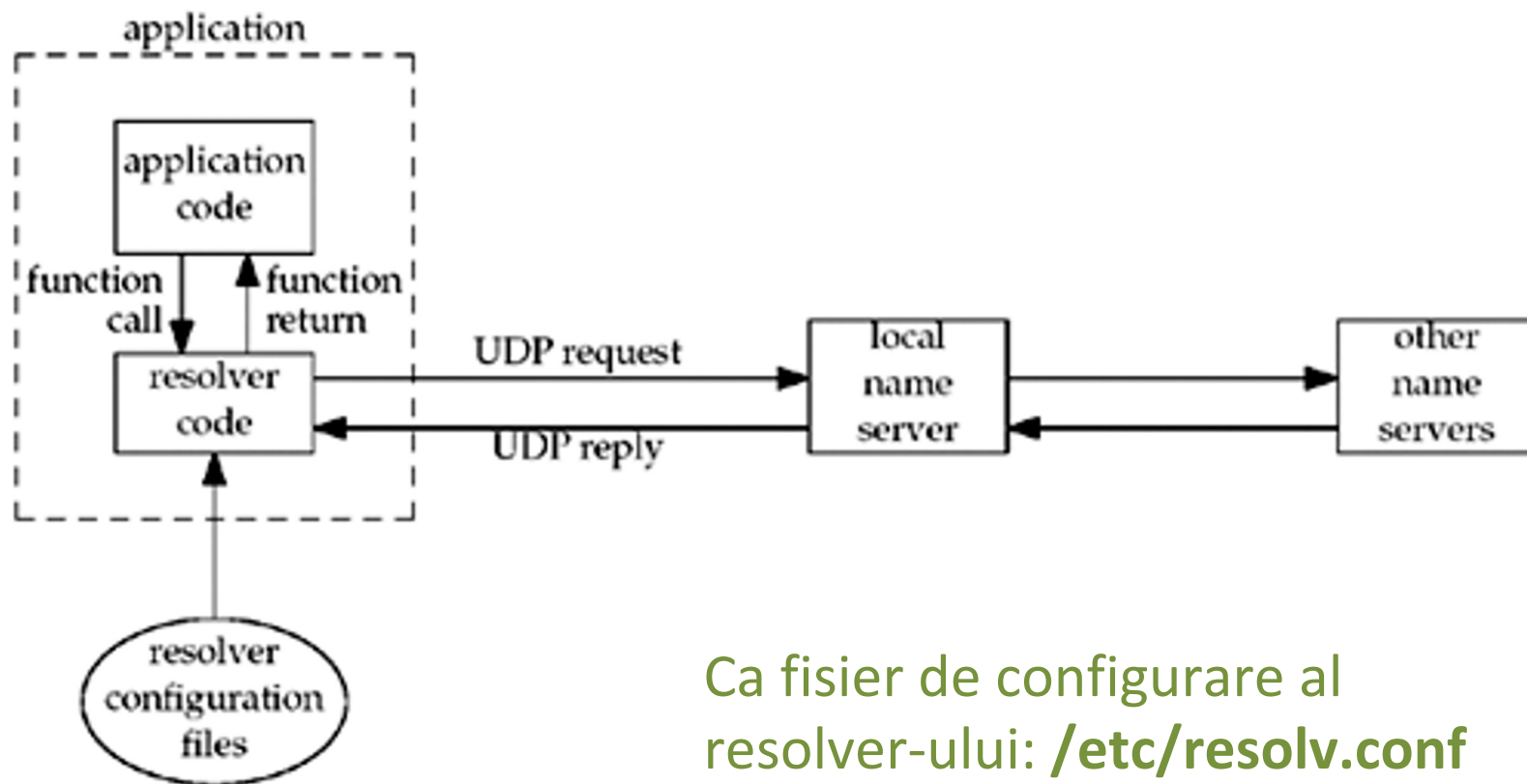
- **SOA** (*Start Of Authority*) – domeniul curent, adresa e-mail a administratorului etc.
- **A** – adresa IP a gazdei (IPv4)
- **AAAA** – adresa IP a gazdei (IPv6)
- **MX** (mail exchangers) – precizeaza numele domeniului pregatit sa accepte posta electronica pentru domeniul specificat
- **CNAME** (*Canonical Name*) – permite crearea pseudonimelor
- **PTR** (*Pointer*) – Pseudonim pentru adresa IP
- **TXT** – text neinterpretat (comentarii)

DNS| configurare

- Exemplu de fisier pentru specificarea unei zone DNS

```
; Zone file for axiologic.ro
;
; The full zone file
;
$TTL 3D
@      IN      SOA      ns1.axiologic.ro. abss.axiologic.ro. (
2007050103      ; serial, todays date + todays serial #
14400           ; refresh, seconds
7200            ; retry, seconds
1209600         ; expire, seconds
1D )            ; minimum, seconds
;
@      IN      NS       ns1.axiologic.ro.      ; Inet Address of name server
@      IN      NS       ns2.axiologic.ro.      ; Inet Address of name server
@      IN      MX       5 mailx.axiologic.ro.   ; Primary Mail Exchanger
;
localhost      A        127.0.0.1
axiologic.ro.  A        72.249.105.153
www            A        72.249.105.153
mailx          CNAME    axiologic.net.
mail           A        207.210.101.144
ftp            A        72.249.105.153
axiologic.ro. IN TXT    "v=spf1 mx mx:mailx.axiologic.ro. ~all"
ns1            A        207.210.101.144
ns2            A        207.210.101.216
~
(END)
```

DNS | clienti, resolveri, servere



Ca fisier de configurare al
resolver-ului: **/etc/resolv.conf**

[Unix Network Programming, 2003,
R. Stevens B. Fenner, A. Rudoff]

DNS| configurare

- Exemplu de fisier /etc/resolv.conf

```
[adria@thor ~] $ cat /etc/resolv.conf
domain info.uaic.ro
search info.uaic.ro
nameserver 85.122.16.1
nameserver 85.122.16.4
[adria@thor ~] $
```

DNS | interogari inverse

- Problema:
 - Data o adresa, care va fi numele ei simbolic?
(*reverse DNS resolution* sau *reverse DNS lookup*)

Exemple:

1)

```
[adria@ns1 ~]$ host 85.122.23.1  
1.23.122.85.in-addr.arpa domain name pointer thor.info.uaic.ro.  
[adria@ns1 ~]$
```

2)

```
2001:db8::567:89ab  
b.a.9.8.7.6.5.0.0.0.0.0.0.0.0.0.0.0.0.0.0.0.8.b.d.0.1.0.0.2.ip6.arpa
```

DNS | optimizari

Proximitatea spatiala: serverele locale vor fi interogate mai des decat cele la distanta

Proximitatea temporală: daca un set de domenii sunt referentiate repetat atunci se apeleaza la *caching DNS*

Fiecare intrare DNS va avea stabilita o valoare TTL (*time to live*)

Se va utiliza si replicarea (servere multiple, servere root multiple) – se va interoga cel mai apropiat (geografic) server

DNS | comenzi

Ca resolver interactiv se pot folosi comenzile:

- **nslookup**
- **dig**
- **host**
- **whois**
- ...

DNS | nslookup

Exemple de utilizari:

\$ nslookup www.info.uaic.ro

- Returneaza RR de tip A folosind serverul DNS local

```
-sh-4.2$ nslookup www.info.uaic.ro
Server:      85.122.16.1
Address:     85.122.16.1#53

Name:   www.info.uaic.ro
Address: 85.122.23.162
```

Host Lookup

\$ nslookup 85.122.23.1

- Returneaza RR de tip PTR pentru 85.122.23.1 in ierarhia de domenii in-addr.arpa

```
-sh-4.2$ nslookup 85.122.23.1
1.23.122.85.in-addr.arpa    name = mail.info.uaic.ro.
1.23.122.85.in-addr.arpa    name = thor.info.uaic.ro.
```

*Reverse IP
Lookup*

[<http://www.zytrax.com/books/dns/ch3/>]

DNS | nslookup

Exemple de utilizari:

\$ nslookup www.axiologic.ro

- Returneaza RR de tip A folosind serverul DNS specificat

```
-sh-4.2$ nslookup www.axiologic.ro 13.248.158.159
Server:          13.248.158.159
Address:         13.248.158.159#53

Name:   www.axiologic.ro
Address: 185.53.177.53
```

Host Lookup

\$ man nslookup

DNS | dig

dig – un instrument mai puternic decat nslookup

Exemplu de
utilizare:

\$ dig www.info.uaic.ro A

```
-sh-4.2$ dig www.info.uaic.ro A

; <<>> DiG 9.11.4-P2-RedHat-9.11.4-26.P2.el7_9.14 <<>> www.info.uaic.ro A
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 7888
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 3, ADDITIONAL: 5

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags::; udp: 4096
;; QUESTION SECTION:
;www.info.uaic.ro.                IN      A

;; ANSWER SECTION:
www.info.uaic.ro.                86400   IN      A      85.122.23.162

;; AUTHORITY SECTION:
info.uaic.ro.                    86400   IN      NS      onix.uaic.ro.
info.uaic.ro.                    86400   IN      NS      nsiasi.roedu.net.
info.uaic.ro.                    86400   IN      NS      orion.uaic.ro.

;; ADDITIONAL SECTION:
nsiasi.roedu.net.                86400   IN      A      192.129.4.100
nsiasi.roedu.net.                86400   IN      AAAA    2001:b30:1:100::100
onix.uaic.ro.                     600     IN      A      85.122.16.4
orion.uaic.ro.                   600     IN      A      85.122.16.1

;; Query time: 0 msec
;; SERVER: 85.122.16.1#53(85.122.16.1)
;; WHEN: Fri Nov 24 22:27:07 EET 2023
;; MSG SIZE rcvd: 207
```

DNS | host

host

Exemplu de utilizare:

```
[~sh-4.2$ host 128.30.52.45  
45.52.30.128.in-addr.arpa domain name pointer blackhole.w3.org.
```


DNS | whois

\$ whois ibm.com

```
Domain Name: ibm.com
Registry Domain ID: 1555443_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.corporatedomains.com
Registrar URL: www.cscprotectsbrands.com
Updated Date: 2023-03-16T01:11:31Z
Creation Date: 1986-03-19T00:00:00Z
Registrar Registration Expiration Date: 2024-03-20T04:00:00Z
Registrar: CSC CORPORATE DOMAINS, INC.
Sponsoring Registrar IANA ID: 299
Registrar Abuse Contact Email: domainabuse@cscglobal.com
Registrar Abuse Contact Phone: +1.8887802723
Domain Status: clientTransferProhibited http://www.icann.org/epp#clientTransferProhibited
Domain Status: serverDeleteProhibited http://www.icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited http://www.icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited http://www.icann.org/epp#serverUpdateProhibited
Registry Registrant ID:
Registrant Name: IBM DNS Admin
Registrant Organization: International Business Machines Corporation
Registrant Street: New Orchard Road
Registrant City: Armonk
```

```
Tech Name: IBM Corporation
Tech Organization: International Business Machines (IBM)
Tech Street: New Orchard Road
Tech City: Armonk
Tech State/Province: NY
Tech Postal Code: 10598
Tech Country: US
Tech Phone: +1.9192544441
Tech Phone Ext:
Tech Fax: +1.9147654370
Tech Fax Ext:
Tech Email: dnstech@us.ibm.com
Name Server: ns1-99.akam.net
Name Server: usc3.akam.net
Name Server: usc2.akam.net
Name Server: eur2.akam.net
Name Server: ns1-206.akam.net
Name Server: asia3.akam.net
Name Server: usw2.akam.net
Name Server: eur5.akam.net
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2023-03-16T01:11:31Z <<<
```

DNS| primitive

- Nu trebuie scris un resolver pentru a afla adresa IP a unei gazde
- Functii principale:
 - `gethostbyname(); gethostbyaddr();` -> *obsolete*
 - `getaddrinfo(); getnameinfo();` -> **recomandate**
- aditionale:
 - `getservbyname(); getservbyport()`

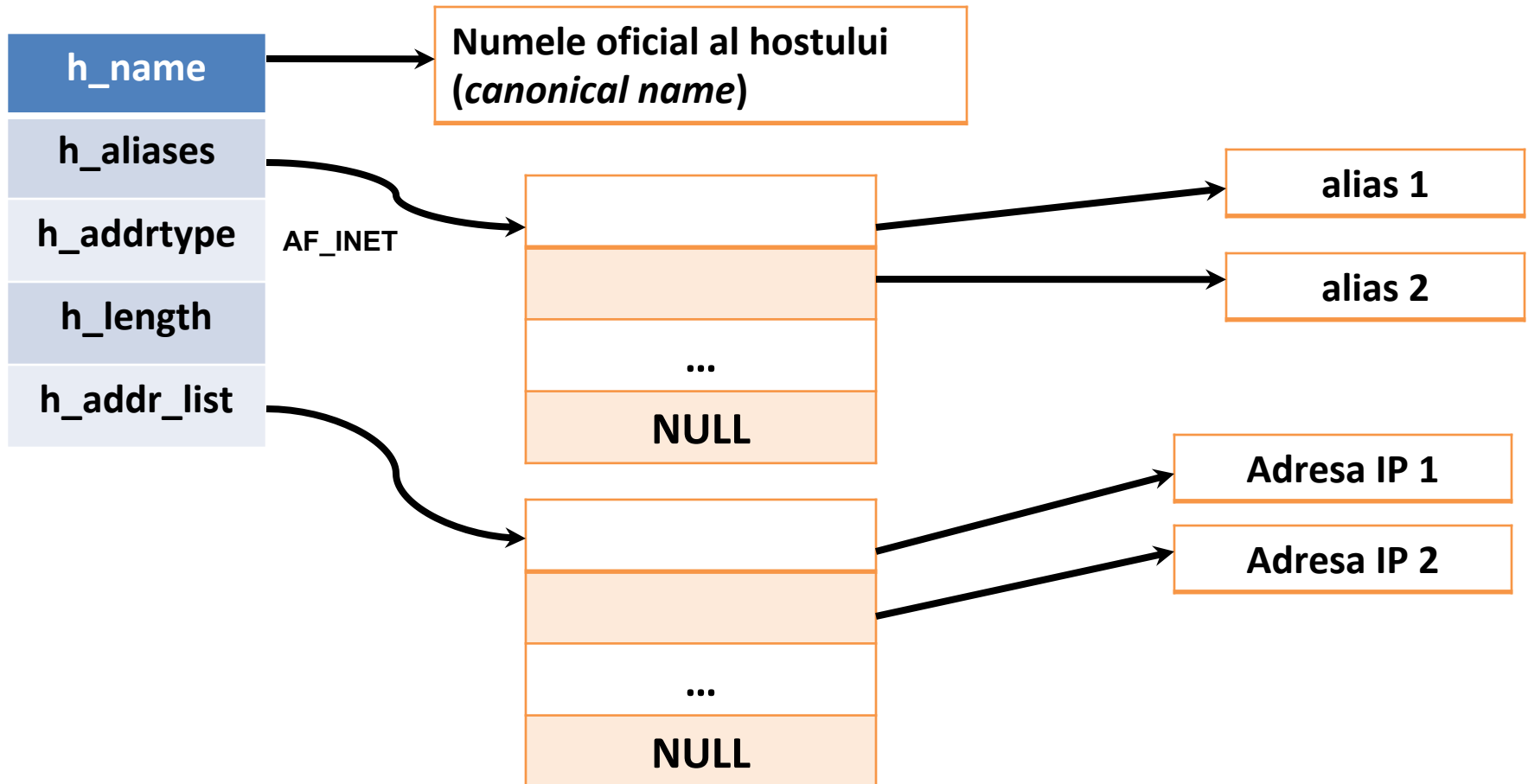
DNS| primitive

Una din structurile folosite: **hostent**

```
struct hostent {  
    char *h_name;      /* nume oficial (canonical) */  
    char **h_aliases; /* alias-uri */  
    int h_addrtype;    /* AF_INET */  
    int h_length;      /* lungimea adresei: 4 sau 6 */  
    char **h_addr_list; /* pointeri la adresele IP */  
};
```

DNS | primitive

Structura **hostent**:



DNS | gethostbyname()

obsolete

```
#include <netdb.h>
```

```
struct hostent *gethostbyname  
                (const char *hostname);
```

- In termenii DNS, `gethostbyname()` realizeaza o cerere pentru o inregistrare **A**
- Obs. `gethostbyname()` se foloseste in special pentru IPv4

DNS | `gethostbyname()`

obsolete

- Returneaza:
 - In caz de succes returneaza un pointer la **hostent**, ce contine adresa IP a *host*-ului
 - In caz de eroare NULL, iar variabila **h_errno** indica eroarea aparuta:
 - **HOST_NOT_FOUND**
 - ...
 - **NO_RECOVERY**
 - ...

} Constante definite in
netdb.h

DNS | `gethostbyname()`

obsolete

- Exemplu de utilizare: completarea structurii *sockaddr_in* avand in loc de adresa IP un nume simbolic:

```
struct sockaddr_in server;  
struct hostent *hos;  
if(!( hos = gethostbyname("fenrir.info.uaic.ro") )  
    { /*Eroare la rezolvarea adresei*/}  
server.sin_family=AF_INET  
    /* adresa IP o luam din structura hos */  
memcpy(&server.sin_addr.s_addr, hos->h_addr_list[0],  
        sizeof(hos->h_addr_list));  
server.sin_port=htons(4321);
```

DNS | `gethostbyaddr()`

obsolete

```
#include <netdb.h>
```

```
struct hostent *gethostbyaddr (  
    const char *addr,  
    socklen_t len,  
    int family);
```

- In termenii DNS, **gethostbyaddr()** realizeaza o cerere la serverul de nume pentru o inregistrare **PTR** in domeniul in-addr.arpa
- Returneaza: In caz de succes returneaza un pointer la **hostent**, ce contine numele oficial al *host*-ului ; In caz de eroare NULL, iar variabila **h_errno** indica eroarea aparuta

Obs. **gethostbyaddr()** se foloseste in special pentru IPv4

DNS | getservbyname()

```
#include <netdb.h>
```

```
struct servent *getservbyname (const char *servname, const char  
    *protoname);
```

- Returneaza: un pointer la *struct servent* in caz de succes, NULL in caz de eroare

```
struct servent {  
    char *s_name;    /* numele oficial al serviciului */  
    char **s_aliases; /* alias-uri */  
    int s_port;      /* portul (network-byte order) */  
    char *s_proto;  /* protocolul */ };
```

Exemplu: struct servent *pserv;

```
pserv=getservbyname("ftp","tcp"); /*FTP folosind TCP */
```

DNS | getservbyport()

```
#include <netdb.h>
```

```
struct servent *getservbyport (int port, const char *protoname);
```

- Cauta un serviciu dupa un numar de port si dupa protocol (optional)
- Returneaza: un pointer la *struct servent* in caz de succes, NULL in caz de eroare

Obs. port este in *network byte order*

Exemplu:

```
struct servent *pserv;
```

```
pserv=getservbyport( htons(53), "udp"); /*DNS folosind UDP */
```

```
pserv=getservbyport( htons(21),"tcp"); /*FTP folosind TCP */
```

DNS | getaddrinfo()

```
#include <netdb.h>
```

```
int getaddrinfo (  
    const char *hostname,  
    const char *service,  
    const struct addrinfo *hints,  
    struct addrinfo **result ) ;
```

Numele host-lui sau o adresa IPv4 sau IPv6 ca string

Portul serviciului sau numele serviciului ("http", "pop", ..) (vezi /etc/services)

Contine informatii despre tipul de informatii pe care trebuie sa le intoarca primitiva

- Obs. *hostname*, *service*, *hints* – parametri de intrare
- Returneaza: 0 in caz de succes, !=0 in caz de eroare
- Se recomanda a fi folosita si pentru IPv4 si pentru IPv6
- Combina functionalitati ale: `gethostbyname()`, `getservbyname()`, `getservbyport()`

DNS | getaddrinfo()

```
struct addrinfo {  
    int ai_flags;          /* AI_PASSIVE, AI_CANONNAME */  
    int ai_family;        /* AF_INET, AF_INET6, AF_UNSPEC */  
    int ai_socktype;      /* SOCK_STREAM sau SOCK_DGRAM */  
    int ai_protocol;     /* 0 (auto) sau IPPROTO_TCP, IPPROTO_UDP */  
    socklen_t ai_addrlen; /* lungimea lui ai_addr */  
    char *ai_canonname;   /* numele canonic al host-ului */  
    struct sockaddr *ai_addr; /* adresa binara a socket-ului */  
    struct addrinfo *ai_next; /* pointer la urmatoarea structura din lista */  
};
```

DNS | getaddrinfo()

Discutii:

- Daca functia returneaza cu succes **result** va pointa la lista de **struct addrinfo**.

Cazuri cind se pot obtine structuri multiple:

- Exista mai multe adrese asociate cu numele hostului si cate o structura este returnata pentru fiecare adresa
- Daca serviciul este furnizat pentru tipuri diferite de *socket*-uri, atunci cate o structura este returnata pentru fiecare tip de *socket*
- Informatia returnata de **getaddrinfo()** in structura **struct addrinfo** ****result** poate fi utilizata astfel:
 - Pentru **socket()** : **ai_family**, **ai_socktype**, **ai_protocol**
 - Pentru **connect()** sau **bind()**: **ai_addr** si **ai_addrlen**
- **freeaddrinfo()**

DNS | getnameinfo()

```
#include <netdb.h>
```

```
int getnameinfo (  
    const struct sockaddr *sockaddr,  
    socklen_t addrlen,  
    char *host,  
    socklen_t hostlen,  
    char *serv,  
    socklen_t servlen,  
    int flags) ;
```

Adresa socket-ului trimisa
ca argument

numele host-ului intors

Numele serviciului

NI_NOFQDN -> **host** va contine
doar numele host-ului si nu intreg
numele al domeniului

- Inlocuieste gethostbyaddr() si getservbyport()
- Returneaza: 0 in caz de succes, !=0 in caz de eroare

DNS | IDN

- **Internationalized Domain Names (IDN)**

- Extensie care permite folosirea caracterelor Unicode in numele de domenii, nu doar a celor ASCII

<https://www.icann.org/resources/pages/idn-2012-02-25-en>

16 Noiembrie 2009 - Inregistrarea de domenii ccIDN (sau IDN ccTLD)

2010-01: ICANN announces that Egypt, the Russian Federation, Saudi Arabia, and the United Arab Emirates were the first countries to have passed the Fast Track String Evaluation within the IDN ccTLD domain application process.

- Pot fi exploatare pentru atacuri de tip *phishing* (detalii intr-un curs viitor)

DNS | administrare

- Radacina DNS este oficial administrata de Internet Corporation for Assigned Names and Numbers (ICANN)
- Exista si alte organizatii care ofera radacini alternative (*alt DNS roots*), precum OpenNIC (Network Information Center), Handshake, sau Yeti DNS

Rezumat

- Domain Name System (DNS)
 - Caracterizare
 - Organizare
 - Configurare
 - Comenzi, Primitive
 - Internationalized Domain Name (IDN)



Intrebari?

Intrebari?